INCH-POUND

MIL-DTL-70729A (AR)
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SUPERSEDING
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31 March 1989

DETAIL SPECIFICATION

CHARGE PROOF, 155 MILLIMETER, PXR-6297A1 LOADING, ASSEMBLING AND PACKING

Reinstated after 25 July 2018 and may be used for new and existing designs and acquisitions.

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1. SCOPE

1.1 Scope. This specification covers the loading, assembling and packing for one type of propelling charge designated as Charge, Proof, PXR-6297A1 for 155mm Cannon, M199/Howitzer, M198.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions or questions on this document should be addressed to: U.S. Army ARDEC, ATTN: RDAR-EIQ-SA, Picatinny Arsenal, NJ 07806-5000 or emailed to ardecstdzn@conus.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database https://assist.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-P-223	Powder, Black
MIL-C-43157	Cloth, Spun Viscose Rayon Resin Impregnated
MIL- A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL- L-48176	Liner (Titanium Dioxide and Paraffin Wax) for Use in Cartridges

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	Identification Marking of US Military Property
MIL-STD-1168	Lot Numbering of Ammunition
MIL-STD-1916	DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available on line at http://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC) DRAWINGS

12598573	Proof Charge, PXR-6297A1, for 155mm Cannon, M199/Howitzer M198
12902708	Packing and Marking for Container, Ammo, Metal PA103 for Charge, Proof, 155mm PXR-6297A1 for Cannon, M199/Howitzer M198

(Copies of these drawings maybe requested by email at usarmy.picatinny.ardec.list.drawing-request-help-desk@mail.mil or from US Army ARDEC, ATTN: RDAR-EIS-PE, Picatinny Arsenal, NJ 07806-5000.)

2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951	Standard Practice for Commercial Packaging

(Copies of ASTM Standards may be ordered online at http://www.astm.org/ or from the ASTM International, 100 Barr Harbor Drive, Conshohocken, PA 19428-2959).

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with drawing 12598573 and MIL-C-70729A.

3.1.1 Liner. The liner shall meet the requirements as specified in Dwg. 9378133 and tested in accordance with MIL-L-48176.

3.2 Propelling charge. The propelling charge shall comply with all requirements specified in drawing 12598573 and MIL-C-70729A.

3.3 Weights.

3.3.1 Propelling weight. The propellant weight shall comply with the weight specified in the loading authorization within plus or minus 0.25 ounce.

3.3.2 Charge, propelling assembly weight. The charge propelling assembly weight shall not differ by more than 2.0 ounces from the total amount specified in the loading authorization plus average weights of the case body, rear cap and igniter assembly, liner assembly, tie cords and tapes.

3.4 Dimensional gage.

3.4.1 Maximum length. The completed charge shall freely enter the prescribed gage. The gage shall be designed to inspect the maximum length.

3.5 Moisture content of black powder. The moisture content of the black powder at the igniter loading station at the time of loading shall not exceed 0.7%. The moisture content of the black powder in the basepad of the propelling charge shall be determined in accordance with MIL-P-223.

3.6 Destructive pull test. The force of the adhesive bond between rear cap and igniter container shall be 200 lbs minimum.

3.7 Proving Ground. Twenty propelling charges will be fired at 70°F for information.

3.7.1 Residue. The propelling charge shall function satisfactorily without excessive or smoldering residue.

3.8 Assembly torque. The test screw shall be assembled to withstand subsequent re-application of the minimum assembly torque specified on drawing 12902708, without moving.

3.9 Workmanship. All parts shall be fabricated and assembled in a thorough, workmanlike manner. They shall be free of dents, cracks, excessive adhesives not cleaned from joints and other foreign material.

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3.10 Ammunition lot numbering. Lot numbers shall be applied per MIL-STD-1168.

4. VERIFICATION

TABLE II. Requirement/verification cross reference matrix

Methods of Verification 1 – Examination 2 – Test				Classes of Verification A- First Article B- Conformance/Acceptance		
Section 3 & 5 Requirements	Description	Verification Method		Verification Class		Section 4 Verification
		1	2	A	B	
3.1	Materials	X	X		x	4.5.9
3.2	Propelling Charge	X	X		X	4.5
3.3	Weights	X			X	4.5.1, 4.5.2, 4.5.3, 4.5.4
3.4	Dimensional gage	X	X		X	
3.5	Moisture content of black powder		X		X	4.5.5
3.6	Destructive pull test		X		X	4.5.6
3.7	Proving ground		X		X	4.5.8
3.8	Assembly torque		X		X	4.5.10
3.9	Workmanship	X			X	4.2.2
5.1	Packaging and packing	X			X	4.2.2.9, 4.2.2.10
5.2	Marking	X			X	4.2.2.11

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Quality conformance inspection.

4.2.1 Inspection lot formation. Inspection lots shall comply with the lot formation provision of MIL-A-48078. In addition, inspection lots of propelling charge assemblies shall contain:

(a) Propellant from not more than one lot number.

(b) Black powder from not more than two lot numbers with the same interfix number from one manufacturer.

(c) CBI powder from not more than two lot numbers with the same interfix number from one manufacturer.

(d) Cases from not more than one interfix number from one manufacturer.

4.2.2 Examination. See MIL-A-48078.

(a) Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables sampling plans for major and minor defects shall be in accordance with MIL-STD-1916.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

4.2.2.1	End, Base Igniter			Drawing Number 9370161
Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Next Higher Assembly 9371062 Inspection Method Reference
Critical	None Defined			
Major				
101	Cloth improper (prior to cutting of basepad) 1/		3.1	4.5.7
102	Diameter of base igniter end	0.40%	3.2	Template
Minor				
201	Evidence of poor workmanship	0.65%	3.9	Visual
NOTES: 1/ See paragraph 4.5.8				

4.2.2.2	Base Igniter Assembly			Drawing Number 9370162
Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Next Higher Assembly 9371063 Inspection Method Reference
Critical	None Defined			
Major				
101	Diameter of inner seam circle	0.40%	3.2	Template
102	Diameter of outer seam circle	0.40%	3.2	Template
103	Inner cavity quilt missing	0.40%	3.2	Visual
Minor				

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201	Tapes not equally spaced (4 places)	0.65%	3.2	Visual
202	Evidence of poor workmanship	0.65%	3.9	Visual
NOTES:				

4.2.2.3	Base Igniter Loading Assembly			Drawing Number 9370163
Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Next Higher Assembly 9345092 Inspection Method Reference
Critical				
1	Assembly incorrect or damaged to the extent that CBI or black powder can escape (including any seams or openings incompletely stitched or broken threads)	100%	3.2	Visual/Manual
Major				
101	Weight of black powder	100%	3.2	4.5.1
102	Weight of CBI	100%	3.2	4.5.2
103	Any seam improper	0.40%	3.2	Visual
104	Moisture content of black powder 1/		3.5	4.5.5
Minor				
201	Evidence of poor workmanship	0.65%	3.9	Visual
NOTES: 1/ See paragraph 4.4.3.3				

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4.2.2.4	Rear Cap and Igniter Assembly			Drawing Number 9345092
Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Next Higher Assembly 12598573 Inspection Method Reference
Critical				
1	Damaged to the extent that black powder or CBI can escape	100%	3.2	Visual
Major				
101	Damaged rear cap (cracks that affect the structural strength and dents that affect the assembly or perforate the material)	100%	3.2	Visual
102	Damaged igniter container (cracks that affect the structural strength and dents that affect the assembly or perforate the material)	100%	3.2	Visual
103	Rear cap and base igniter loading assembly not properly assembled to the igniter container, adhesive not visible for majority of circumference, more than 1/8 in. of tape end (4 places) exposed outside the assembly	100%	3.2	Visual
104	Destructive pull test 1/		3.6	4.5.6
Minor				
201	Evidence of poor workmanship	0.65%	3.9	Visual
NOTES: 1/ See paragraph 4.4.3.2				

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4.2.2.5	Lead Laminated Rayon Cloth			Drawing Number 9370160
Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Next Higher Assembly 9345104 Inspection Method Reference
Critical	None defined			
Major				
101	Established length	0.40%	3.2	Scale
Minor				
201	Evidence of poor workmanship	0.65%	3.9	Visual
NOTES:				

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4.2.2.6	Propellant and Liner Assembly			Drawing Number 12910074
Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Next Higher Assembly 12598573 Inspection Method Reference
Critical	None defined			
Major				
101	Propellant not properly bundled and not tied tightly (2 places of lacing cords and 1 place of the tape prior to liner assembly assembled)	0.40%	3.2	Visual
102	Propellant weight (prior to assembly of liner) 1/	100%	3.3.1	See 4.5.3
103	Liner assembly improper, improperly assembled, missing or improperly located, lead surface not facing propellant and middle lacing cord loose or not abutted against end of liner	100%	3.2	Visual/Scale
Minor				
201	Evidence of poor workmanship	0.65%	3.9	Visual
NOTES: 1/ One stick or propellant may be allowed to break up to achieve the propellant weight tolerance and the broken stick shall be put in the middle of the propellant bundle				

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4.2.2.7	Proof, Charge, PXR-6297A1 for 155MM Cannon M199/Howitzer M198			Drawing Number 12598573
Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Next Higher Assembly Inspection Method Reference
Critical				
1	Base igniter loading assembly missing, damaged or improper to the extent that black powder or CBI can escape (to be inspected from inside of the rear cap and igniter assembly and before assembly to the body)	1/	3.2	Visual-Manual
2	Base igniter loading assembly damage or improper to the extent that black powder or CBI can escape (to be inspected from outside of assembly and after assembled to the body)	100%	3.2	Visual-Manual
Special	Proving Ground			
a	Pressure	20 samples	3.7	4.5.8.1
Major				
101	Damaged case (cracks that affect the structural strength and dents that affect the assembly or perforate the material)	100%	3.2	Visual
102	Rear cap and igniter assembly not properly snapped in, bosses not completely inside the holes	100%	3.2	Visual
103	Frayed buttons (after assembly button damaged to the extent that button becomes soft and cotton-like materials are sticking out) See 6.17	100%	3.2	Visual

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104	End of bundle with liner not inserted first (prior to assembly of rear cap and igniter assembly)	100%	3.2	Visual
105	Dimensional gage failure	100%	3.4.1	Gage

4.2.2.7	CONTINUATION SHEET			Drawing Number 12598573
Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Next Higher Assembly Inspection Method Reference
106	Propelling charge assembly weight	100%	3.3.2	4.5.4
Minor				
201	Markings missing, misleading or unidentifiable	0.65%	3.2	Visual
202	Evidence of poor workmanship	0.65%	3.9	Visual
Informational				
301	Moisture content of black powder 1/		3.5	4.5.5
NOTES: 1/ One basepad per eight (8) hour shifts				

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4.2.2.8	Strap and Filler Assembly			Drawing Number 9362602/9362603
Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Next Higher Assembly Inspection Method Reference
Critical	None defined			
Major				
101	Diameter of filler (Dwg 9362603-1)	0.40%	3.2	Gage
Minor				
201	Strap not centrally located (Dwg 9362603)	0.65%	3.2	Visual
202	Chain-off length incorrect (Dwg 9362603) if applicable	0.65%	3.2	Scale
203	Minimum strap length (Dwg 9362602)	0.65%	3.2	Scale
204	Strap width (Dwg 9362602) 1/	0.65%	3.2	Scale
205	Marking missing, misleading or unidentifiable (Dwg 9362603)	0.65%	3.2	Visual
206	Evidence of poor workmanship	0.65%	3.9	Visual

NOTES:
1/ If a slitting machine is utilized, in lieu of the specified AQL, during the slitting process each roll will be inspected within the first 10 feet and the last 10 feet; i.e., two measurements per roll

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4.2.2.9	Container, Prior to Insertion of Charge			Drawing Number 12902708
Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Next Higher Assembly Inspection Method Reference
Critical	None defined			
Major				
101	Foreign matter or rust in container	100%	3.2	Visual
102	Bottom cushion missing or improper (type or thickness)	100%	3.2	Visual
103	Damaged container (to include dents protruding greater than 1/4" into container)	100%	3.2	Visual – Gage (see 6.4)
Minor	None defined			
NOTES:				

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4.2.2.10	Container, Prior to Sealing			Drawing Number 12902708
Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Next Higher Assembly Inspection Method Reference
Critical				
1	Warning label (part no. 12912974) missing	100%	3.2	Visual
Major				
101	Any component missing or improper	0.40%	3.2	Visual
102	Dimension “X”, minimum (in lieu of measuring Dimension “X”, the measurement of Dimension “Y”, maximum may be taken)	100%	3.2	Scale
103	Filler or strap missing, filler incorrect type, filler on the top of strap, or wrapper above the top of charge	100%	3.2	Visual
104	Top cushion missing or incorrect type	100%	3.2	Visual-Scale
105	Top spacer (if applicable) incorrect type	100%	3.2	Visual
Minor	None defined			
NOTES:				

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4.2.2.11	Container, Sealed			Drawing Number 12902708
Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Next Higher Assembly Inspection Method Reference
Critical	None defined			
Major				
101	Cover incompletely engaged	0.40%	3.2	Visual
102	Test screw missing	100%	3.2	Visual
103	DOD symbol missing, misleading or unidentifiable (if applicable)	0.40%	3.2	Visual
104	Test screw assembly torque	0.40%	3.8	Test (see 6.12)
105	Air pressure test	100%	3.2	4.5.10
Minor				
201	Metallic seal missing or incorrectly applied	0.65%	3.2	Visual
202	Marking missing, misleading or unidentifiable	0.65%	3.2	Visual
NOTES:				

4.2.2.12	Containerized Charge			Drawing Number 12902708
Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Next Higher Assembly Inspection Method Reference
1/	Tear down inspection 2/		3/	4.5.9
NOTES: 1/ Category is Critical or Major dependent upon the category of the defect inspected 2/ See paragraph 4.4.3.1 3/ Requirement paragraph is dependent upon inspection performed				

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4.3 Testing

4.3.1 Teardown testing. Five (5) completely assembled charges shall be randomly selected from the first three consecutive lots and tested in accordance with 4.5.10.

4.3.2 Destructive pull test. Five (5) rear cap and igniter assemblies, drawing 9345092, shall be the same charges used from the teardown inspection and tested in accordance with 4.5.6.

4.3.3 Moisture content of black powder. At least one sample of black powder from each eight hours loading of black powder spot assemblies will be tested for moisture content.

4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and 6.3 of this specification.

4.5 Methods of inspection.

4.5.1 Weight of black powder. The inspections below are to be conducted by inspection personnel using equipment designated for inspection purposes and shall be clearly distinguished from production operations. The weight of the black powder will be inspected 100 percent. The weight of the black powder in the spot will be determined by subtracting an average weight of the empty assembly (no CBI or black powder) or the actual weight as marked on the individual assembly from the loaded assembly weight.

4.5.2 Weight of CBI. The inspections below are to be conducted by inspection personnel using equipment designated for inspection purposes and shall be clearly distinguished from production operations. The weight of the CBI will be inspected 100 percent. The weight of the CBI in the basepad will be determined by subtracting an average weight of the empty assembly (no CBI) or the actual weight as marked on the individual assembly from the loaded assembly weight.

4.5.3 Propellant weight. The inspections are to be conducted by inspection personnel using equipment designated for inspection purposes and shall be clearly distinguished from production operations. Any propelling charge which fails to comply with the applicable weight requirement (see 3.3) shall be classed defective and removed from the lot. Inspection equipment shall be in accordance with 4.4.4. The propellant bundle (without liner assembly) will be inspected 100 percent. The weight of the propellant will be determined by subtracting an average weight of the tapes from the propellant bundle (without liner assembly).

4.5.4 Charge, propelling, assembly weight. The average weights of the case body, rear cap and igniter assembly, liner assembly, cords and tapes, together with the nominal propellant weight as specified in the loading authorization shall be subtracted from the assembled charge, propelling, assembly weight. If the resulting weight deviation is greater than the requirement, the propelling charge will be classified defective and removed from the lot (see 6.11).

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4.5.5 Moisture content of black powder. The moisture content of the black powder will be determined in accordance with MIL-P-223. If any sample fails to meet the specified requirement, the eight hours production represented by the sample shall be rejected. Subsequent samples taken from basepad at time of charge assembly will be for informational purpose.

4.5.6 Destructive pull test. This test shall be performed after the adhesive bond between the igniter container and rear cap has had sufficient time to set. The assembly shall be held in a suitable test fixture clamping on the igniter container. The test apparatus shall apply a longitudinal load distributed over 360o around the lip of the rear cap. Observation shall be for the force required for separation of the rear cap from the igniter container. If any sample fails to meet the specified requirement, the lot shall be rejected.

4.5.7 Cloth, spun viscose rayon resin impregnated. Each roll, as a minimum, shall be tested for compliance with the physical characteristics listed in Table I in 3.5 of MIL-C-43157. Testing will be in accordance with 4.3.1 of MIL-C-43157. Any sample which fails to comply with the applicable requirement shall be cause for rejection of the cloth bolt represented by the sample. The method of inspection to be used during production is to be established by the contractor and approved by the government. The method, as a minimum, shall contain positive assurance that the material being utilized has been verified for compliance with the specification requirements as stated above (see 6.14).

4.5.8 Proving ground assessments. These tests shall be conducted at a government proving ground in accordance with the applicable acceptance test procedure using M549, RA projectiles inert loaded within 0.1 pound of an as fired weight of 96.0 pounds or M864 projectiles (103.0 lbs) using primer, M82 and fired from M198 Howitzer with M199 with 95 percent (minimum) remaining life. Firing shall be conducted with a spindle to charge base stand-off of 1 inch (plus or minus 1/4 inch). Prior to firing, all charges shall be temperature conditioned in a large conditioning box with adequate air circulation at the required temperature for a minimum of 24 hours and assurance will be made that, when fired, the charges are at the conditioning temperature. The following tolerances shall apply 70°F conditioning: 70°F from 68°F to 72°F. Copper crusher gages (2) shall be used to measure the maximum chamber pressure of each charge. Velocity coils or radars will be used to measure the muzzle velocity of each charge.

4.5.8.1 Pressure. Two pressure-time gages at different locations (one in the spindle and one at the projectile base) will be used to record pressure-time traces. Differential pressure traces will be obtained by subtracting the projectile gage output from the spindle gage output. The maximum differential pressure of any charge will be defined as the maximum amplitude of the first negative wave of the differential pressure trace prior to peak pressure. Any trace which contains no amplitude in the negative direction will be considered to have a differential pressure of zero. Ignition delay will be recorded on all differential pressure firings. Twenty (20) charges will be conditioned to 70°F and ten (10) charges will be fired with the M549 (inert loaded) and ten (10) with the M864 (inert). Ten (10) calibration charges will be fired intermixed (five (5) with each projectile) with the test samples and the muzzle velocity and pressure of all test charges corrected to standard conditions.

4.5.8.2 Residue (see 6.9). All propelling charges fired at different temperatures in accordance with 4.5.8.1 shall be observed for excessive residue and smoldering residue. A test lot shall be referred to the procuring activities with a copy of test data forwarded to the technical agency for disposition whenever any of the following conditions exist:

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(a) Flaming, smoking or smoldering residue in a stationary position on the breech surface, chamber, or tube surface after the firing pulse.

(b) Evidence of two or more rounds exhibiting excessive residue (see 6.9.1).

4.5.9 Teardown inspection. Each container and propelling charge will be subjected to each of the critical (except special defect a, 4.2.2.7) and 100 percent major (except major defect 101, major defect 102, 4.2.2.3; major defect 102, major defect 103, 4.2.2.6; major defect 105, major defect 106, 4.2.2.7; major defect 102, 4.2.2.10) inspections defined in 4.2.2 of this specification. Samples may be accumulated or inspected upon selection provided the selection remains random and represents the lot. The inspections will be performed by the contractor and witnessed by the Government. The detection of any one of the above defects shall be cause for rejection of the lot represented by the sample.

4.5.10 Air pressure test. Each container shall be tested as specified on drawing 12902708. A pressure drop equivalent to or greater than the applicable requirement shall be cause for rejection and removal from the lot.

5. PACKAGING

5.1 Packing

5.1.1 Level A. The propelling charges shall be packed, sealed and tested in accordance with Dwg 12902708.

5.1.2 Level B. Same as specified in 5.1.1.

5.1.3 Level C. Same as specified in 5.1.1.

5.2 Marking

5.2.1 Level A, B & C. Mark in accordance with Dwg 12902708.

6. NOTES

6.1 Intended use. The proof charges covered by this specification are intended for use with 155MM projectiles, development and acceptance tests, and for M199 cannon proof tests. This proof charge is for Proving Ground use only, to be fired in a M199 cannon with all personnel behind a protective barrier.

6.2 Ordering data. See MIL-A-48078

6.3 Submission of designs for approval. See 6.2.3 of MIL-A-48078. Submit equipment designs, as required, to Commander, US Army Armament Research, Development and Engineering Center, ATTN: RDAR-EIQ-MA, Picatinny Arsenal, New Jersey 07806-5000. This address will be specified on the Contract Data Requirements List DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

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6.4 Visual examination qualification. The tape should be stored in the original container in a cool location. It should not be stored close to steam pipes, radiators or other sources of heat, or in locations where it may be exposed to the direct rays of the sun.

6.5 Inspection lot size. It is noted that the size of inspection lots of components, assemblies, or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests of deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

6.6 Proving Ground. A copy of all proving ground data should be submitted to Commander, US Army Armament Research, Development and Engineering Center, ATTN: RDAR-EIQ-MA and RDAR-MEE-W, Picatinny Arsenal, New Jersey 07806-5000.

6.7 Charge manufacture. All test and calibration charges should be manufactured in accordance with this specification.

6.7.1 Test charge. The test charges should be loaded by the proving ground and the proving ground will furnish a complete set of operating procedures (including inspections to be performed and gage drawings) for approval to the technical agency prior to any manufacturing of test charges.

6.7.2 Calibration lot. All calibration lots should meet the lot formation criteria of 4.2.1 of this specification except no more than one lot of CBI or BP should be used. Because of the non-availability of long term storage containers (metal) for stick propellant and combustible case components, and in order to maintain homogeneity, the calibration lot should be loaded by the contractor responsible for LAP of this charge in production. The entire lot will be loaded and sealed in PA103 containers at the same time (no further proving ground testing is required to accept the calibration lot). However, charges needed for the establishment testing of calibration lot may be loaded at the proving ground.

6.8 Data cards. A copy of all data cards should be submitted to Commander, US Army Armament Research, Development and Engineering Center, ATTN: RDAR-EIQ-MA and RDAR-MEE-W, Picatinny Arsenal, New Jersey 07806-5000.

6.9 Residue

6.9.1 Excessive residue. Excessive residue should be defined as residue large enough to prevent chambering or otherwise hinder functioning of a subsequent round. In addition, any single piece of residue larger than 1 inch by 1 inch should be considered excessive. Any collected residue and photographs should be forwarded to Commander, US Army Armament Research, Development and Engineering Center, ATTN: RDAR-MEE-W, Picatinny Arsenal, New Jersey 07806-5000 for evaluation.

6.10 Teardown inspection. A list of all defects found during teardown inspection should be submitted to Commander, US Army Armament Research, Development and Engineering Center, ATTN: RDAR-EIQ-MA, Picatinny Arsenal, New Jersey 07806-5000.

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6.11 Charge, propelling, assembly weight.

Weights to Be Determined	Assembly Drawing Number
Average weight of body	9345098
Average weight of rear cap and igniter assembly	9345092
Average weight of liner assembly	9378133
Average weight of the tapes and cords	9345104
Nominal propellant weight	Loading Authorization
Component weight (sum of above weights)	

The propelling charge assembly weight minus the component weight equals the “weight deviation”.

6.12 Assembly torque. The assembly torque of the test screw will apply at time of loading only due to the relaxation of the gasket material.

6.13 Lot formation. Allowance for two black powder or CBI lots is for the purpose to facilitate LAP when a black powder or CBI lot is depleted and a new lot is introduced.

6.14 Method of inspection submission. Methods of inspection for resin impregnated spun viscose rayon cloth should be submitted to Commander, US Army Armament Research, Development and Engineering Center, ATTN: RDAR-EIQ-MA, Picatinny Arsenal, New Jersey 07806-5000 for approval.

6.15 Automated inspections or processes. When components or assemblies are inspected or produced by automated methods or punches or slitting machines where variations are known to be minimal, the contract may submit a description of the process and process controls along with proposed new or reduced sampling for evaluation and approval to Commander, US Army Armament Research, Development and Engineering Center, ATTN: RDAR-EIQ-MA, Picatinny Arsenal, New Jersey 07806-5000.

6.16 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, ARRADCOM or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.17 Rework on frayed button. When the propelling charge is rejected because of frayed button, rework may be performed on the frayed button for re-accepting the propelling charge. The contractor should submit a description of the rework procedure to QA COR for evaluation and approval and with a copy furnished to Commander, RDAR-EIQ-MA, Picatinny Arsenal, New Jersey 07806-5000.

6.18 Subject term (key work) listing.

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Artillery
Cannon
Proof Charge
Howitzer
Projectiles

6.19 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army-AR

Preparing Activity:
Army-AR

Project: 1320-2018-008

NOTE: the activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https:/assist.dla.mil.

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